DDA3020 Machine Learning

Tutorial: Decision Trees and Random Forests

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Instructions

This tutorial covers the main ideas of decision trees and random forests, including basic tree terminology, entropy and information gain, tree construction, overfitting, bagging, random forests, and the difference between classification trees and regression trees.

Show your working clearly for all calculation questions. You may use a calculator for $\log_2$ values unless otherwise stated.

Useful Formulas

For a node S with class proportions $p_1, p_2, \ldots, p_K$,

$$\text{Entropy}(S) = -\sum_{i=1}^{K} p_i \log_2 p_i.$$

For binary classification with positive-class proportion p,

$$\text{Entropy}(S) = -p \log_2 p - (1-p)\log_2(1-p).$$

If an attribute A splits node S into subsets $D_1, D_2, \ldots, D_V$, then

$$\text{Entropy}(S \mid A) = \sum_{v=1}^{V} \frac{|D_v|}{|S|} \text{Entropy}(D_v),$$

and

$$\text{Gain}(A) = \text{Entropy}(S) - \text{Entropy}(S \mid A).$$

Exercises

Q1. Which of the following statements about a decision tree is correct?

(A) A decision tree is a parametric model because it learns parameters from data.

(B) A decision tree is often considered interpretable because its prediction follows an explicit decision path.

(C) A decision tree can only be used for classification, not regression.

(D) A decision tree is usually trained by finding the global optimum tree exactly.

Q2. In a decision tree, a leaf node usually stores

(A) the next splitting rule,
(B) the final prediction,
(C) the information gain of the root,
(D) the depth of the whole tree.

Q3. If all samples in a node belong to the same class, the entropy of that node is

(A) 1,
(B) 0.5,
(C) 0,
(D) impossible to determine.

Q4. If a feature has the largest information gain, this usually means that

(A) it produces child nodes with lower impurity,
(B) it must be a continuous feature,
(C) it must have the largest number of possible values,
(D) it must be linearly related to the label.

Q5. What is the key difference between bagging and random forest?

(A) Bagging uses many trees, while random forest uses only one tree.
(B) Bagging is only for regression, while random forest is only for classification.
(C) In random forest, each split considers only a random subset of features.
(D) Random forest does not use bootstrap sampling.

Q6. Which of the following is most likely to cause overfitting in a single decision tree?

(A) Using a very small maximum depth,
(B) Growing the tree very deep until nodes are nearly pure,
(C) Using majority voting across many trees,
(D) Using feature subsampling.

Q7. A node contains 10 samples: 8 positive and 2 negative. Compute the entropy of this node.

Q8. A node contains 6 samples, and all 6 belong to the same class. Compute the entropy of this node.

Q9. A parent node S contains 8 samples: 4 positive and 4 negative, so

$$\text{Entropy}(S) = 1.$$

A feature A splits S into two child nodes:

- S_1: 4 samples, with 3 positive and 1 negative;
- S_2: 4 samples, with 1 positive and 3 negative.

Compute

(a) Entropy(S_1),
(b) Entropy(S_2),
(c) Entropy($S \mid A$),
(d) Gain(A).

Q10. A parent node contains 12 samples: 6 positive and 6 negative, so

$$\text{Entropy}(S) = 1.$$

Consider the following two candidate splits.

Split A

- Left child: 6 positive, 0 negative
- Right child: 0 positive, 6 negative

Split B

- Left child: 4 positive, 2 negative
- Right child: 2 positive, 4 negative

Which split is better according to information gain? Briefly explain why.

Q11. In a binary decision tree, the longest path from the root node to a leaf node contains 4 edges. What is the depth of the tree?

Q12. A decision tree has 1 root node, 3 internal nodes excluding the root, and 5 leaf nodes. What is the size of the tree?

Q13. Describe the basic top-down greedy learning procedure of a decision tree in a few clear steps.

Q14. Why is a decision tree usually considered more interpretable than logistic regression?

Q15. Why can a single deep decision tree easily overfit the training data?

Q16. Why is random forest often more powerful than plain bagging of decision trees?

Q17. What is the main difference between a classification tree and a regression tree?

Q18. For binary classification, the entropy of a node is

$$H(p) = -p \log_2 p - (1-p) \log_2 (1-p),$$

where p is the proportion of positive samples. Explain why the entropy is largest when $p = 0.5$ and smallest when $p = 0$ or $p = 1$.

Q19. Bagging often works very well for decision trees. Explain why bagging is especially effective for high-variance base learners.

Q20. A student says: *"Random forest must be more interpretable than a single decision tree, because it is more accurate."* Do you agree? Explain clearly.

Q21. Consider a dataset with only two features. One feature is extremely strong and almost always gives a very good split near the root.

(a) Why might many bagged trees still look very similar?
(b) Why can random forest help more in this case?

Q22. What may happen if the maximum depth of a decision tree is set too small or too large? Relate your answer to underfitting and overfitting.

Solutions

Q1. Answer: B.

Explanation: A decision tree is generally regarded as a nonparametric model. Its main advantage is interpretability: for a given prediction, we can follow the path from the root to a leaf and see each decision made along the way. Option A is incorrect because trees are not usually described as parametric models. Option C is incorrect because trees can be used for both classification and regression. Option D is incorrect because standard tree learning is greedy rather than an exact global optimization procedure.

Q2. Answer: B.

Explanation: A leaf node represents the end of the decision process and stores the final output, such as a class label in classification or a numerical value in regression. It does not store a further splitting rule.

Q3. Answer: C.

Explanation: If all samples belong to the same class, the node is pure. A pure node has no uncertainty, so its entropy is 0.

Q4. Answer: A.

Explanation: Information gain measures how much uncertainty is reduced after a split. If a feature gives the largest information gain, it means that the resulting child nodes are, on average, more pure than before.

Q5. Answer: C.

Explanation: Bagging trains multiple trees on bootstrap samples. Random forest does that as well, but adds another source of randomness: at each split, only a random subset of features is considered. This helps reduce correlation among trees.

Q6. Answer: B.

Explanation: A very deep tree can keep splitting until the leaves are almost pure, which allows it to fit not only the real pattern but also noise in the training data. This increases the risk of overfitting.

Q7. Answer:

$$H = -0.8 \log_2(0.8) - 0.2 \log_2(0.2) \approx 0.722.$$

Explanation: The node contains 8 positive and 2 negative samples, so the class proportions are 0.8 and 0.2. Substituting these into the entropy formula gives approximately 0.722. Since the node is not pure, the entropy is greater than 0.

Q8. Answer:

$$H = -1 \log_2(1) = 0.$$

Explanation: All samples belong to the same class, so the node is pure. A pure node has no class uncertainty, so its entropy is 0.

Q9. Answer:

$$\text{Entropy}(S_1) = -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4} \approx 0.811,$$

$$\text{Entropy}(S_2) = -\frac{1}{4}\log_2\frac{1}{4} - \frac{3}{4}\log_2\frac{3}{4} \approx 0.811,$$

$$\text{Entropy}(S \mid A) = \frac{4}{8}(0.811) + \frac{4}{8}(0.811) = 0.811,$$
$$\text{Gain}(A) = 1 - 0.811 = 0.189.$$

Explanation: Both child nodes have the same class distribution, namely $3 : 1$ or $1 : 3$, so they have the same entropy. The conditional entropy is the weighted average of the child-node entropies. Since the parent entropy is 1, subtracting the conditional entropy gives the information gain.

Q10. Answer: Split A is better.

Explanation: In Split A, both child nodes are pure, so each child has entropy 0. Therefore, the conditional entropy after the split is 0, and the information gain is 1. In Split B, both child nodes still contain a mixture of classes, so their entropies are positive. As a result, the conditional entropy is greater than 0 and the information gain is smaller. Therefore Split A is the better split.

Q11. Answer: 4.

Explanation: The depth of a tree is the number of edges on the longest path from the root to a leaf. Since the longest path contains 4 edges, the depth is 4.

Q12. Answer:

$$1 + 3 + 5 = 9.$$

Explanation: The size of a tree is the total number of nodes. Here there is 1 root node, 3 internal nodes, and 5 leaf nodes, so the total is 9.

Q13. Answer: A typical answer should mention the following process: start with all training samples at the root; check whether a stopping condition is satisfied; if not, choose the best split according to a criterion such as information gain; partition the data into child nodes; then repeat the same process recursively for each child node.

Explanation: This is called a top-down greedy learning procedure because the tree is built from the root downward, and at each step the algorithm makes a locally best choice rather than searching for a globally optimal tree.

Q14. Answer: A decision tree is usually more interpretable because its prediction is made through an explicit sequence of if-then decisions.

Explanation: For a particular sample, we can trace one path from the root to a leaf and explain exactly why the sample receives its prediction. Logistic regression is also interpretable to some extent through its coefficients, but its prediction process is less direct and usually less intuitive for beginners.

Q15. Answer: A single deep decision tree can easily overfit because it may keep splitting until the training samples are separated almost perfectly.

Explanation: This makes the tree very flexible. While such flexibility can reduce training error, it may also make the model learn noise or accidental patterns in the training set, which hurts generalization on unseen data.

Q16. Answer: Random forest is often more powerful because it reduces both the variance of individual trees and the correlation among trees.

Explanation: Bagging already reduces variance by training trees on bootstrap samples. However, if some features are very strong, many trees may still choose similar top splits. Random forest addresses this by restricting each split to a random subset of features, which encourages diversity among trees and improves the ensemble effect.

Q17. Answer: A classification tree predicts a discrete class label, while a regression tree predicts a continuous numerical value.

Explanation: Because the tasks are different, their splitting criteria are also different. Classification trees usually use measures such as entropy or Gini impurity, whereas regression trees usually reduce squared error or variance.

Q18. Answer: The entropy is largest at $p = 0.5$ and smallest at $p = 0$ or $p = 1$.

Explanation: Entropy measures uncertainty. When $p = 0$ or $p = 1$, all samples belong to one class, so there is no uncertainty and the entropy is 0. When $p = 0.5$, the two classes are equally mixed, so uncertainty is greatest, and entropy reaches its maximum value.

Q19. Answer: Bagging is especially effective for high-variance base learners because their predictions change a lot when the training data changes slightly.

Explanation: Decision trees are a typical high-variance model. By training many trees on bootstrap samples and averaging their predictions, bagging stabilizes the result and reduces variance, which usually improves generalization.

Q20. Answer: No, I do not agree.

Explanation: Higher accuracy does not necessarily mean higher interpretability. A single decision tree is usually more interpretable because it provides one explicit decision path. A random forest combines predictions from many trees, so although it is often more accurate, it is usually harder to explain directly.

Q21. Answer:

(a) Many bagged trees may still look similar because bootstrap sampling changes the data only slightly, and the very strong feature may still be selected near the root in many trees.

(b) Random forest helps more because it forces each split to consider only a random subset of features, so not every tree can rely on the same strong feature at every split.

Explanation: The main goal of random forest is to make the trees less correlated. If all trees repeatedly use the same dominant feature, averaging them brings less improvement. Feature subsampling increases diversity and usually strengthens the ensemble.

Q22. Answer: If the maximum depth is too small, the tree may underfit. If it is too large, the tree may overfit.

Explanation: A shallow tree may be too simple to capture the true structure of the data, which leads to high bias and underfitting. A very deep tree may become overly complex and fit noise in the training set, which leads to high variance and overfitting.